Information system acquisition, development and Implementation

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- IT auditors must understand how project management tools and techniques are used to manage the risks associated with IS acquisition, development, and implementation, and should be able to evaluate if projects are progressing to plan and are on track to deliver the benefits contained in the business case

Business Case Development and Approval

- Business case is an important consideration in any IT project, whether the development of a new system or investment in new infrastructure.
- A business case provides the information required for an organization to decide whether a project should proceed
- The initial business case normally derives from a feasibility study as part of project planning.
- The feasibility study will normally scope the problem, identify and explore a number of solutions and make recommendations on what action to take.
- The business case should be of sufficient detail to describe the justification for setting up and continuing a project

Business case development

- First Step in Project Commencement
- Any IT Project should be driven by a business case i.e the direct link to a business benefit.
- Initially it is derived out of a feasibility study, it scopes the problems, identifies the solution& provide recommendations.
- Also the business case should continue to exist throughout the life of the project- else should exit at “kill points” or “stage gates” (i.e various project progress review points

Business Case Development and Approval

- The feasibility study will normally include the following six elements:
  - **Project Scope** – defines the business problem and/or opportunity to be addressed
  - **Current Analysis** – defines and establishes an understanding of a system, a software Product. At this point in the process, the strengths and weaknesses of the current system or software product are identified.
  - **Requirements** – defined based upon stakeholder needs and constraints
  - **Approach** – Recommended system and/or software solution to satisfy the Requirements
  - **Evaluation** – is based upon the previously completed elements within the feasibility study. The final report addresses the cost-effectiveness of the approach selected
  - **Review** – A formal review of feasibility study report is conducted with all stakeholders

Project Management- General aspects

- IS projects may be initiated from any part of an organization
- A project is always a time-bound effort
- Project management should be a business process of a project-oriented organization
- Project management process begins with the project charter and ends with completion of the project.
- The complexity of project management requires a careful and explicit design of the project management process

Project Communication and culture

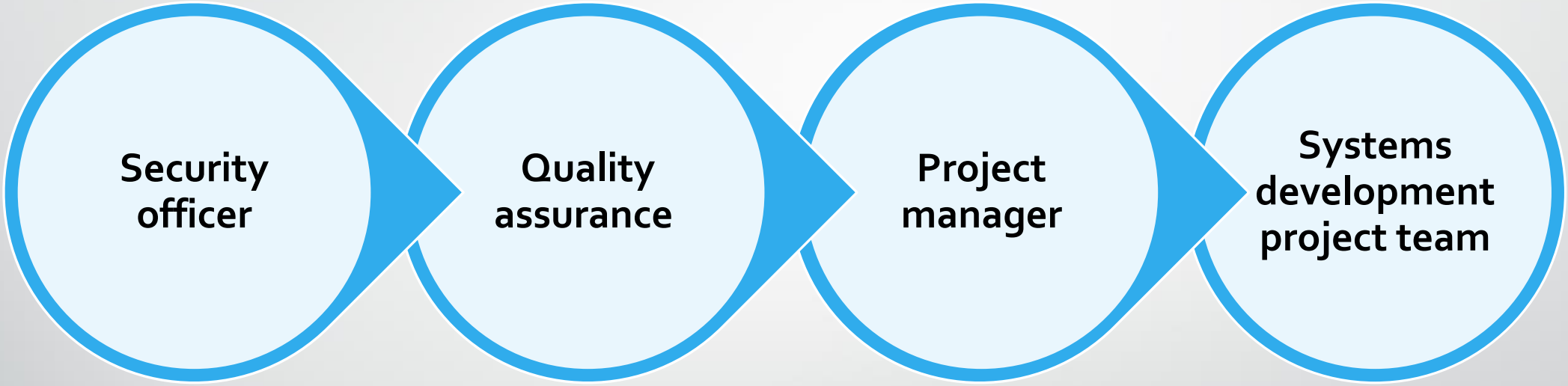
- Project communication can be achieved by:-
  - **One-on-one meetings** – One-on-one meetings and a project start workshop help to facilitate two-way communication between the project team members and the project manager
  - **Kick-off meetings** – A kick-off meeting may be used by the project manager to inform the team of what has to be done for the project
  - **Project start workshops** – communication is open and clear among the project team to use a project start workshop to obtain cooperation from all team members and buy-in from stakeholders. This helps develop a common overview of the project and communicates the project culture early in the project.
  - **A combination of the three**
- A project culture is comprised of shared norms, beliefs, values and assumptions of the project team.
- A key success factor for establishing the correct project culture is defining and adapting the unique characteristics of a project

Roles and Responsibilities



- Demonstrates commitments to the project and approve the resources
- Assumes ownership of the project and resulting systems, allocates qualified resources, and actively participates in business process redesign, system requirement definitions, test case development, acceptance testing and user training
- It provides overall directions and also responsible for all deliverables, project cost, and schedules
- Providing funding for the project.
- Provides technical supports for hardware and software environment by developing, installing User project team – Completes assigned task, communicates effectively with user by actively involving them in the development process as a subject matter expert.

Roles and Responsibilities



- Ensures that systems controls and supporting processes provide an effective level of protection based on data classifications
- Person who review results and deliverables within each phase and at the end of each phase and confirm compliance requirement and operating the requested systems.
- Day to day management and leadership of the project
- Completes assigned task, communicates effectively with user by actively involving them in the development process.

Project Planning- objectives

- Project objectives are the specific action statements that support the road map to obtain established project goals.
- A project needs clearly defined results that are specific, measurable, attainable, realistic and timely (SMART)

Project Planning- objectives

- These objectives are broken down into three
  - **Main objectives** are the primary reason for the project and will always be directly coupled with business success.
  - **Additional objectives** are objectives that are not directly related to the main results of the project but may contribute to project success.
  - **Non-objectives** are the results that are not to be expected on completion of the project.
- A commonly accepted approach to define project objectives is to start off with an object breakdown structure (OBS).
- After the OBS has been compiled or a solution is defined, a work breakdown structure (WBS) is designed to structure all the tasks that are necessary to build up the elements of the OBS during the project

Project Planning- Objectives

OBS – Object based Structure

- It represents the individual components of the solution and their relationships to each other in a hierarchical manner, either graphically or in a table.
- An OBS can help, especially when dealing with non-tangible project results such as organizational development, to ensure that a material deliverable is not overlooked.

Project Planning- Objectives

Work Breakdown Structure

- WBS is designed to structure all the tasks that are necessary to build up the elements of the OBS during the project.
- The WBS represents the project in terms of manageable and controllable units of work, serves as a central communications tool in the project, and forms the baseline for cost and resource planning.

Project Planning

The project manager needs to determine:

- The various tasks that need to be performed to produce the expected business application system
- The sequence or the order in which these tasks need to be performed
- The duration or the time window for each task
- The priority of each task
- The IT resources that are available and required to perform these tasks
- Budget or costing for each of these tasks
- Source and means of funding

Project Planning

- Scheduling and establishing the time frame
- Involves establishing the sequential relationship among tasks
- The schedule can be graphically represented using various techniques such as **Gantt charts** or Program Evaluation Review Technique (**PERT**) diagram

Gantt Charts

- A Gantt chart is a type of bar chart that illustrates a project schedule, named after its popularizer, Henry Gantt (1861–1919), who designed such a chart around the years 1910–1915.
- Modern Gantt charts also show the dependency relationships between activities and the current schedule status.

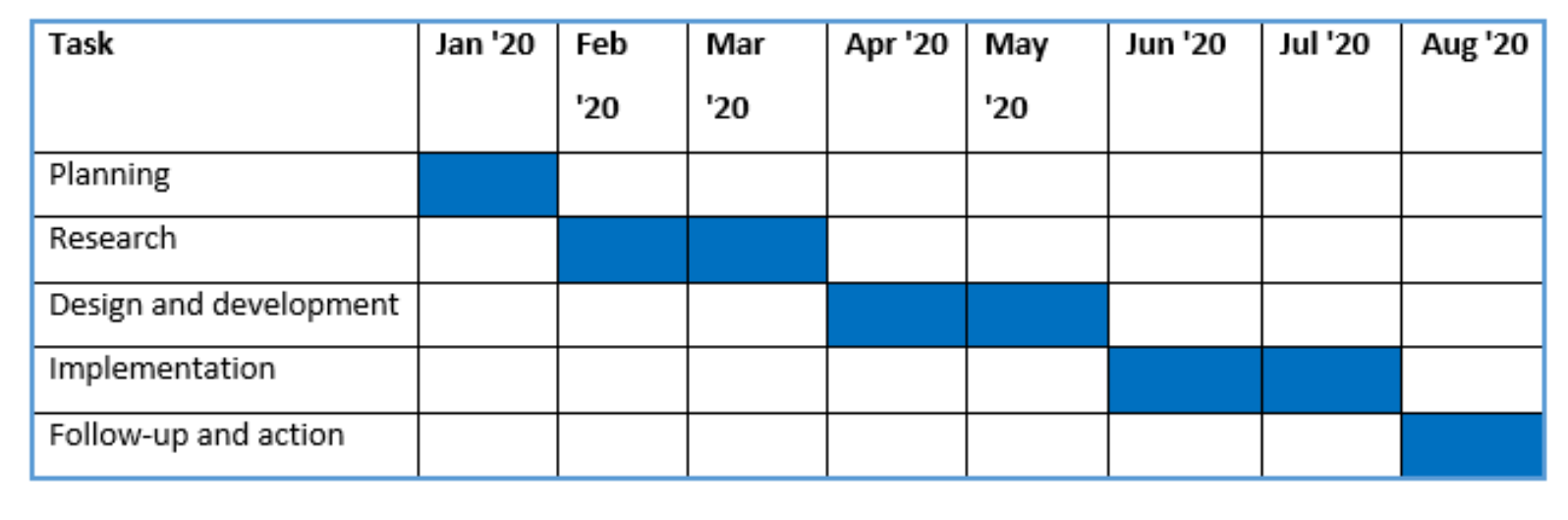
Gantt Charts

- A Gantt chart is a project management tool that assists in planning and scheduling projects of all sizes, although they are beneficial for simplifying complex projects.
- This chart lists the tasks to be performed on the vertical axis and time intervals on the horizontal axis. The width of the horizontal bars in the graph shows the duration of each activity.
- This chart is constructed to aid in scheduling the activities (tasks) needed to complete a project

Gantt Charts

- Gantt charts can also be used to track the achievement of milestones or significant accomplishments for the project, such as the end of a project phase or completion of a key deliverable.
- Progress of the entire project can be read from the Gantt chart to determine whether the project is behind, ahead, or on schedule compared to the baseline project plan
- The advantage of using a Gantt chart is that it helps you monitor your project's progress and set priorities.
- The disadvantage of the Gantt chart is that it does not show the dependency of tasks to each other, and it does not show which tasks are 'critical' to finishing the project on time.

Gantt Charts



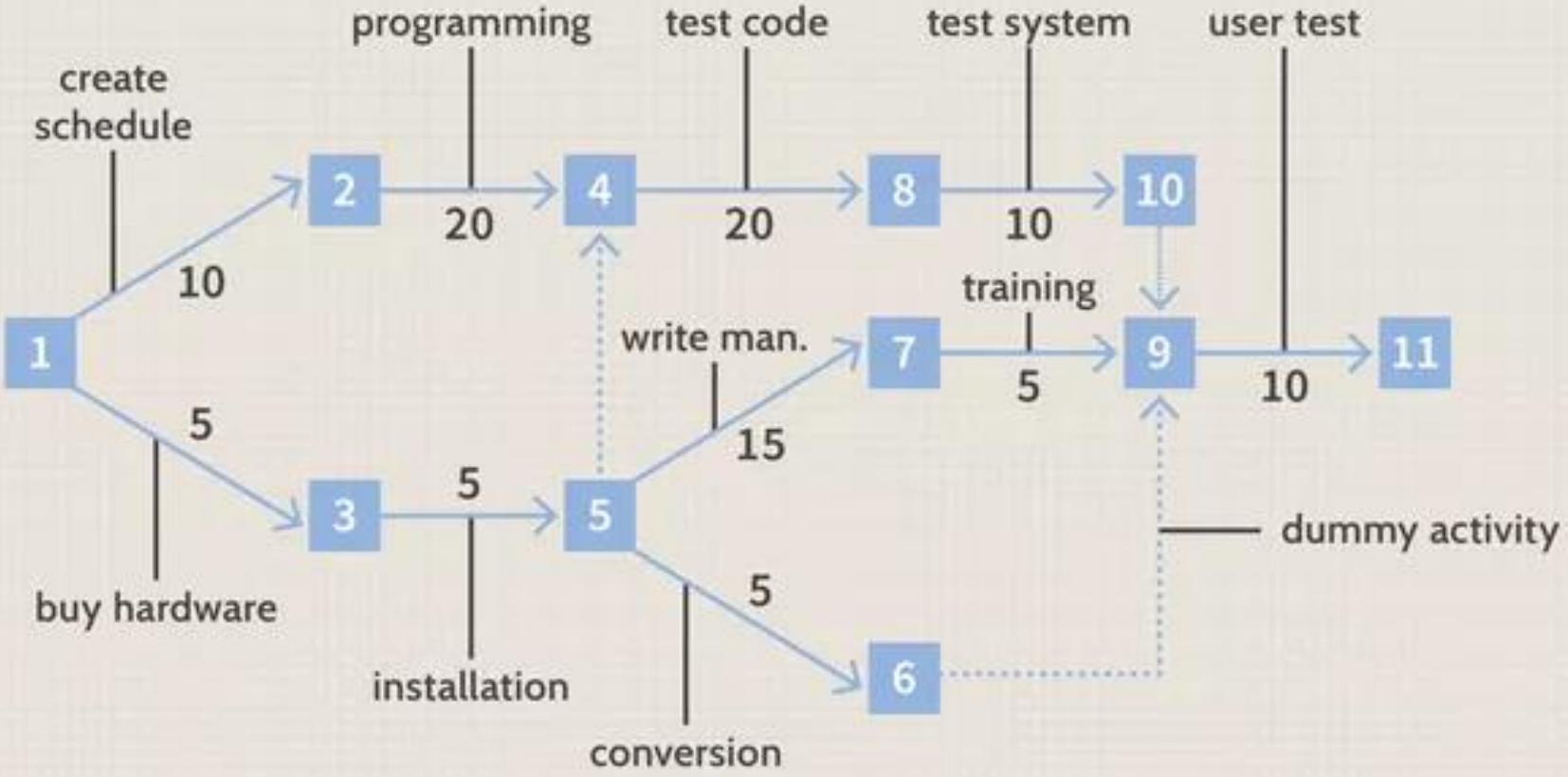
Task	Jan '20	Feb '20	Mar '20	Apr '20	May '20	Jun '20	Jul '20	Aug '20
Planning								
Research								
Design and development								
Implementation								
Follow-up and action								

Program Evaluation Review Technique (PERT)

- The Program Evaluation Review Technique, or PERT, is a visual tool used in project planning.
- Using the technique helps project planners identify start and end dates, as well as interim required tasks and timelines.
- The information is displayed as a network in chart form.

Program Evaluation Review Technique (PERT)

- PERT has a set series of steps in mapping out a complex project, which include:
  - List all the tasks and milestones (a.k.a. events) required for completion of the project
  - Determine the required sequence of tasks
  - Design a chart to visually display all the steps
  - Estimate the time required for each task
  - Identify the critical path – the longest series of tasks in the project
  - Adjust the chart to reflect progress made once the project starts



- Numbered rectangles are nodes and represent events or milestones.
- Directional arrows represent dependent tasks that must be completed sequentially.
- Diverging arrow directions (e.g. 1-2 & 1-3) indicate possibly concurrent tasks.
- Dotted lines indicate dependent tasks that do not require resources.

Business Application Development

- Business applications are very important to organizations as they enhance and monitor productivity throughout the organization.
- These business applications can either be bought as off-the-shelf solution or can be built in-house.
- For applications that are developed in-house, they follow particular system Development life cycle.
- Organizations can choose to implement Software Development Life cycle(SDLC)

SDLC

- Also referred to as the waterfall technique
- Traditional system Development Life Cycle Approach

- **What are the phases in the SDLC?**

SDLC - Phases

- **Phase 1 – Feasibility Study:**
  - Includes development of a *business case*, which determine the strategic benefits of implementing the system either in productivity gains or in future cost avoidance
  - Intangible factors such as readiness of the business users and maturity of the business processes will also be considered and assessed.
  - This business case provides the justification for proceeding to the next phase.
- **Phase 2 – Requirements definition –** Define the problem or need that requires resolution and define the functional and quality requirements of the solution system
- **Phase 3A – Software selection and acquisition (Purchased systems) –** Based on requirements defined, prepare a request for proposal outlining the entity requirements to invite bids from suppliers
- **Phase 3B – Design (In-house development) –** Based on the requirements defined, establish a baseline of system and subsystem specifications that describe the parts of the system, how they interface, and how the system will be implemented using the chosen hardware, software and network facilities.

SDLC Phases

- **Phase 4A – Configuration (purchased systems) –** Configure the system, if it is a packaged system, to tailor it to the organization's requirements. This is best done through the configuration of system control parameters, rather than changing program code.
- **Phase 4B – Development (In-house development) –** Use the design specifications to begin programming and formalizing supporting operational processes of the System
- **Phase 5 – Final testing and implementation –** The system also may go through a certification and accreditation process to assess the effectiveness of the business application in mitigating risk
- **Phase 6 – Post implementation –** Following the successful implementation of a new or extensively modified system, implement a formal process that assesses the adequacy of the system and projected cost benefit or ROI measurements vis-à-vis the feasibility stage findings and deviations

Approaches of test plans

Bottom-up approach:

- a testing strategy in which the modules at the lower level are tested with higher modules until all the modules and aspects of the software are tested properly
- **Benefits of bottom-up approach:**
  - No need for stubs or drivers
  - Can be started before all programs are complete
  - Errors in critical modules are found early

Top-down approach:

- High-level modules are tested first and then low-level modules and finally integrating the low-level modules to a high level to ensure the system is working as intended.
- **Benefits of top-down approach:**
  - Tests of major functions and processing are conducted early
  - Interface errors can be detected sooner
  - Confidence is raised in the system because programmers and users actually see a working system

TESTING

- Testing is an essential part of the development process that verifies and validates that a program , subsystem or application preforms the function which it has been designed.
- Also determines whether the units being tested operate without malfunction or adverse effect on other components of the system.
- Each set of test is performed with different data and under the responsibility of different people or function.
- The IS auditor play a preventative role in testing ensuring that all major controls are included in the system.

What are the various testing classifications?

- Unit Testing
- System testing
- Integration/interface testing
- Final acceptance testing – QAT & UA

Testing classifications

- **Unit testing**
  - The testing of an individual program or module.
  - Unit testing uses a set of test cases that focus on the control structure of the procedural design.
  - These tests ensure that the internal operation of the program performs according to specification.
- **Interface or integration testing**
  - The tests that verify and validate the functioning of the application under test with other systems, where a set of data is transferred from one system to another
  - A hardware or software test that evaluates the connection of two or more components that pass information from one area to another
  - The objective is to take unit-tested modules and build an integrated structure dictated by design.

Testing classifications

- **System testing:**
  - Series of tests designed to ensure the modified programs, objects, database schema, which collectively constitute a new or modified system, function properly.
  - The testing of the software application as a whole to check if the system is complaint with the user requirements.
  - It is an end to end user perspective testing intended to find defects in the software system.

Testing classifications

- **Final acceptance testing:**
  - After the system staff is satisfied with their system tests, the new or modified system is ready for the acceptance testing, which occurs during the implementation phase.
  - Final acceptance testing has two major parts:
  - **Quality assurance testing (QAT):**
    - QAT focuses on the documented specifications and the technology employed.
    - QAT is performed primarily by the IT department.
    - The participation of the end user is minimal and on request.
    - QAT does not focus on functionality testing.
  - **User acceptance testing (UAT):**
    - UAT should be performed in a secure testing or staging environment
    - On completion of acceptance testing, the final step is usually a certification and accreditation process

What are the other types of testing?

Alpha and beta testing	• Early version of an application released to internal users or testing and might not contain all the features planned for in the final version.
Functional testing	• used to test functionality of the system against the detailed requirements to ensure that the software has been built according to user requirements.
White box testing	• Assesses the effectiveness of software programming logic. A method used to test a software taking into consideration its internal functioning. • Test data are used in determining accuracy of the program
Parallel testing	• Parallel Testing is a process to leverage automation testing capabilities by allowing the execution of the same tests simultaneously in multiple environments, real device combinations, and browser configurations.
Black box testing	• The Black Box Test is a test that only considers the external behavior of the system; the internal workings of the software is not taken into account.

Change over or Go live techniques

- Changeover refers to an approach to shift users from using the application from the old system to new system.
- This is appropriate only after testing the new system with respect to its program and relevant data.

Change over or Go live techniques

- **Parallel changeover:**
  - This technique includes running the old system, then running both the old and new systems in parallel, and finally, fully changing over to the new system after gaining confidence in the working of the new system.
  - **Advantages:**
    - minimize the risk of using the newer system
    - help in identifying problems, issues or any concerns that the user comes across in the newer system in the beginning
  - **Disadvantages:**
    - running two systems at the same time is higher costs.
    - The parallel changeover process also can be quite time-consuming.

Change over or Go live techniques

- **Phased changeover:**
  - The phased changeover technique is considered a compromise between parallel and direct changeovers.
  - In a phased changeover, the new system is implemented one stage at a time
  - **Advantages:**
    - Low cost
    - Isolates errors
  - **Disadvantages:**
    - the process takes a long time to complete because phases need to be implemented separately.

Change over or Go live techniques

- **Abrupt changeover:**
  - In this approach the newer system is changed over from the older system on a cutoff date and time, and the older system is discontinued once changeover to the new system takes place
    - **Advantages**:
      - Low cost
    - **Disadvantages**:
      - Asset safeguarding
      - Data integrity
      - System effectiveness
      - System efficiency
      - Change management challenges (depending on the configuration items considered)
      - Duplicate or missing records (duplicate or erroneous records may exist if data cleansing is not done correctly)

Risks with Software development

- New system might not meet user requirements
- Project activities exceed the limits of the financial resources set aside
- Inadequate project management
- Scope creep
- Project team members leaving

Auditing systems development, acquisition and maintenance

- IS auditors tasks in system development acquisition and maintenance may take place once the project is finished or infrequently-during the project itself.
  - Meet with key system development and project team members to determine the main components, objectives and user requirements of the system in order to identify areas that need control.
  - Discuss the selection of appropriate controls with systems development and user project team members to determine and rank the major risks to exposures of system
  - Evaluate available controls and participate in discussions with system development and user project team members regarding

Auditing systems development, acquisition and maintenance

- The IS auditor will participate in post implementation reviews to:
  - Discuss and examine supporting records to test system maintenance procedures to ensure that they are being applied as to standards
  - Identify and test existing controls to determine the adequacy of production library security to ensure the integrity of the production resources.

Auditing systems development, acquisition and maintenance

- Post implementation review
  - Determine if the systems objectives and requirements were achieved.
  - Determine if the cost benefits identified in the feasibility study are being measured , analyzed and accurately reported to management.
  - Review controls built into the system to ensure that they are operating according to design .
  - Review operator error logs to determine if they any problems inherent within the system.
  - Review input and output control balances and reports to verify that the system is processing data accurately.